MEDICAL ROBOTICS PROJECT

Surgical Workflow Analysis

Temporal phase recognition on Cholec80

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THE CLINICAL NEED

Why temporal segmentation, not classification.

A context-aware system must know the current phase before it can:

- Provide intraoperative guidance
- Index and retrieve surgical video
- Anticipate the next surgical step



Similar Phases

Adjacent phases can look alike in a single isolated frame



Visual Noise

Smoke, blood, lighting changes, and tool occlusion

Frame-level noise breaks the logical order of the workflow

Clinical context: Lalys & Jannin (2014); Padoy (2019); Project report, Chapter 1

DATASET DESIGN

Cholec80 becomes a strict video-level benchmark

80
laparoscopic procedures

7
surgical phases

1 fps
from 25 fps video

184,578
aligned frames

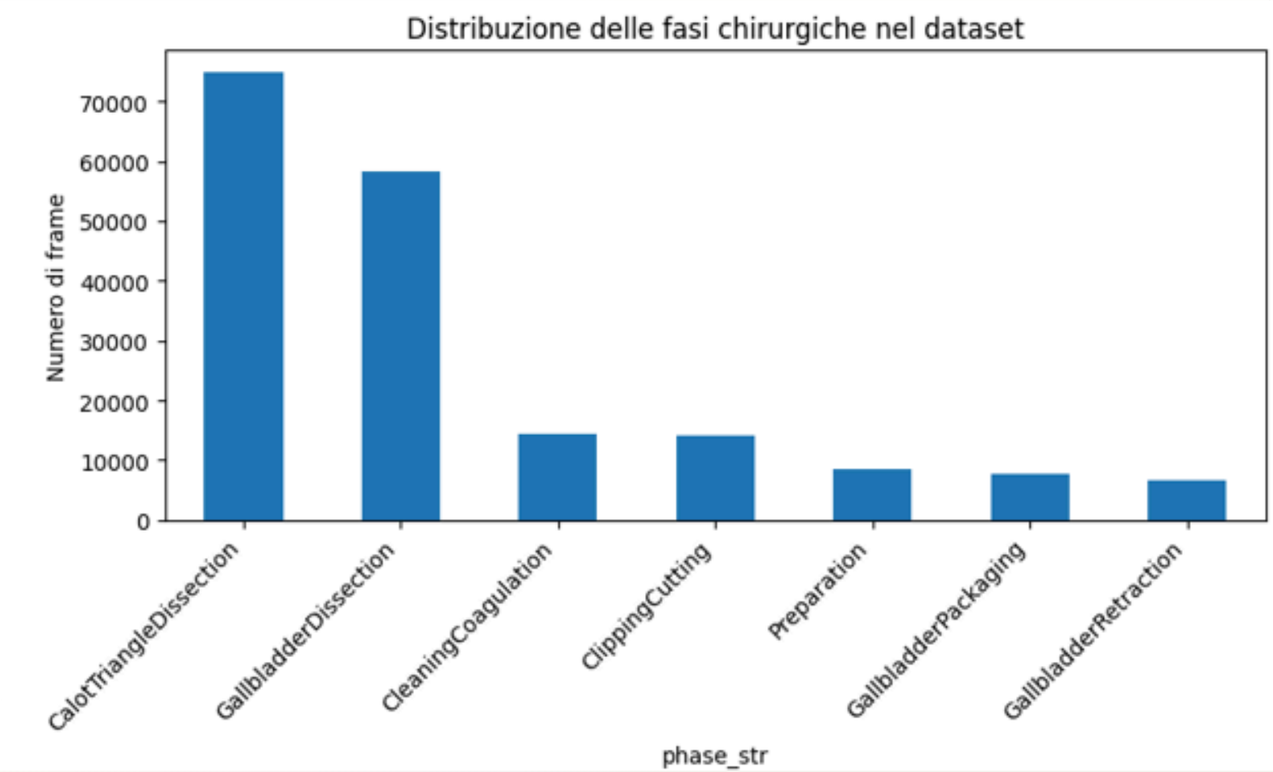


No frame-level split: entire videos stay in one partition to prevent leakage.

Source: Project report, Sections 2.2-2.4 · Cholec80 introduced by Twinanda et al. (EndoNet, 2017)

THE DATA CHALLENGE

Real surgical duration creates an 11× class imbalance



72%+
of frames belong to the two dominant dissection phases.

3.6%
of frames represent gallbladder retraction, the rarest class.

Verified
missing opening or closing phases reflect partial recordings, not annotation failure.

Source: Project report, Section 2.5

A CONTROLLED MODEL INTERFACE

The temporal models share one input–output interface

Video at 1 FPS → **ResNet50** → **T × 2048 features** → **Temporal model** → **T × 7 logits**

teammates
Frozen ImageNet ResNet50 · frames cached as JPEG

original
Frozen ImageNet ResNet50 · direct video decoding

finetuned
Layer 3 + layer 4 adapted to the laparoscopic domain

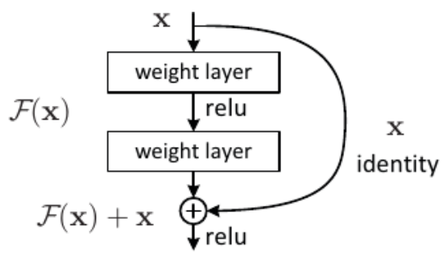
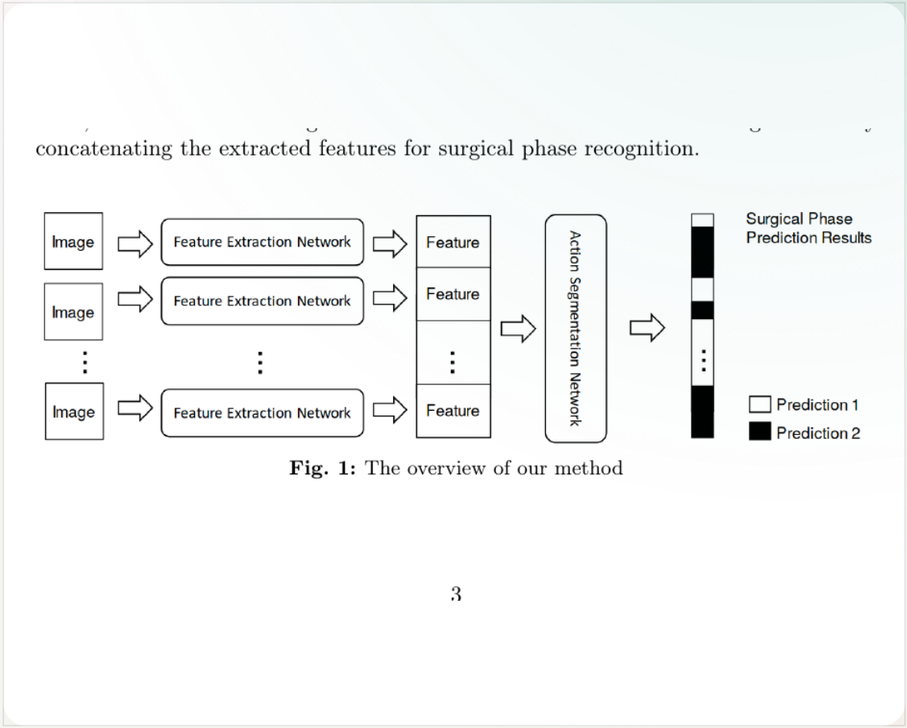


Figure 2. Residual learning: a building block.



Controlled comparison: features change the spatial representation; architectures change temporal reasoning.

Figures: He et al., CVPR 2016, Fig. 2 · Zhang et al., IJCARS 2024, Fig. 1

Memory and attention see the surgical timeline differently

Shared recurrent projection: 2048 → 256 → LayerNorm → GELU → dropout

GRU	**Past only.** Two 256-unit layers compress preceding frames into one hidden state.
BiGRU	**Past + future.** 128 units per direction; 256 dimensions after concatenation.
MS-CBiGRU	**Local + global.** Dilated CNN spans 3 / 5 / 9 / 17 s before bidirectional recurrence.

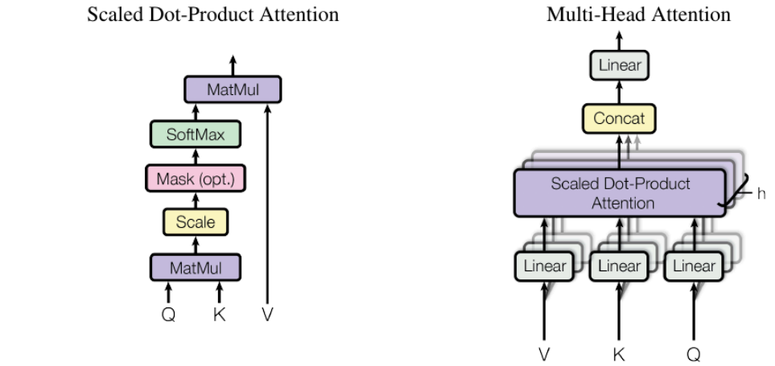


Figure 2: (left) Scaled Dot-Product Attention. (right) Multi-Head Attention consists of several attention layers running in parallel.

3 layers · 8 heads

Positional encoding inside overlapping 64 s windows; overlap logits are averaged at inference.

Attention figure: Vaswani et al., NeurIPS 2017, Fig. 2 · Recurrent specifications: project notebooks

ARCHITECTURE DEEP DIVE

MS-TCN sees the whole video and refines it four times

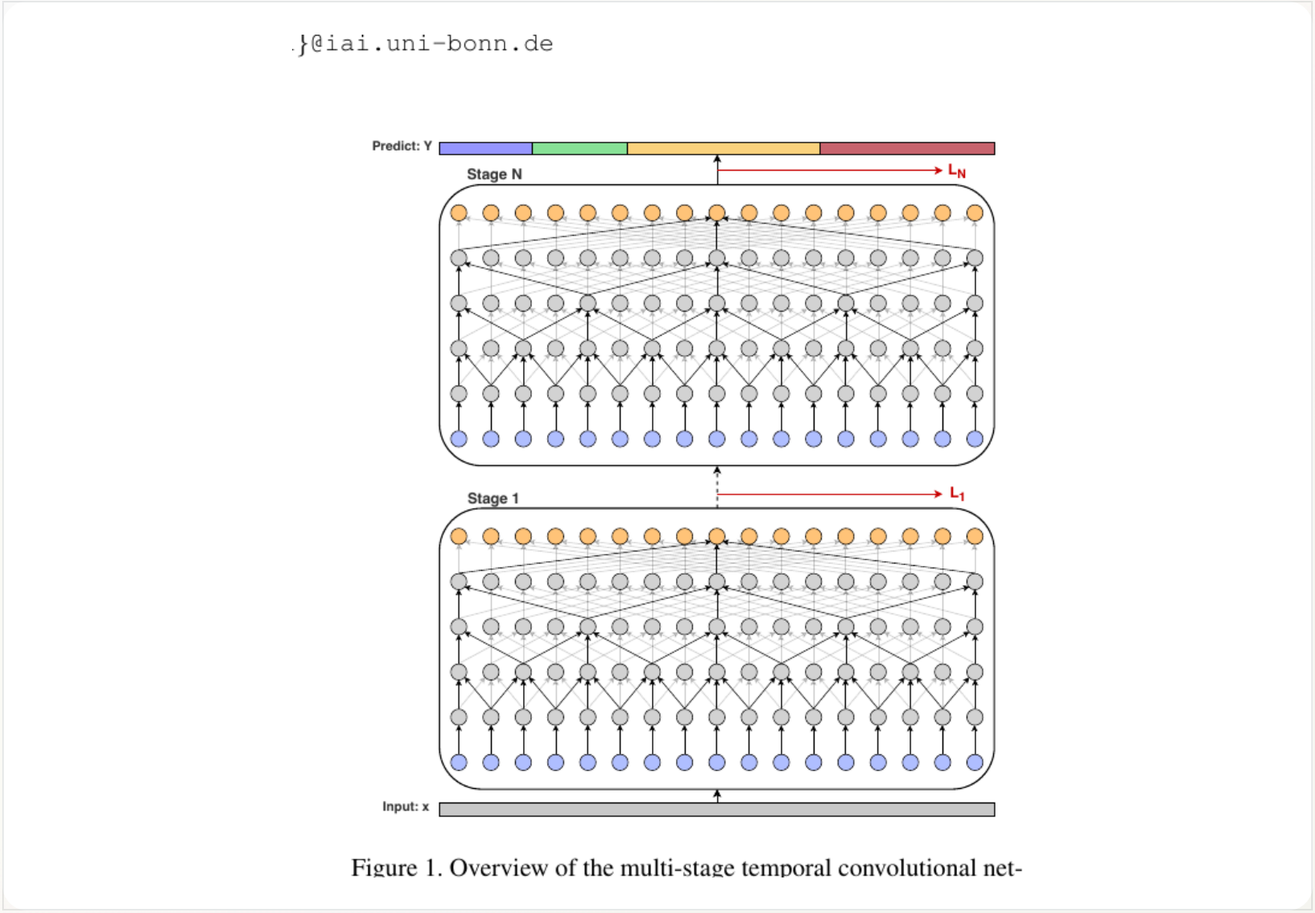


Figure 1. Overview of the multi-stage temporal convolutional net-

Full sequence

2048 input channels are projected to 64 temporal channels.

2,047 / 4,095 s

Nominal context from 10 / 11 exponentially dilated layers.

4 supervised stages

Initial prediction followed by three probability refinements.

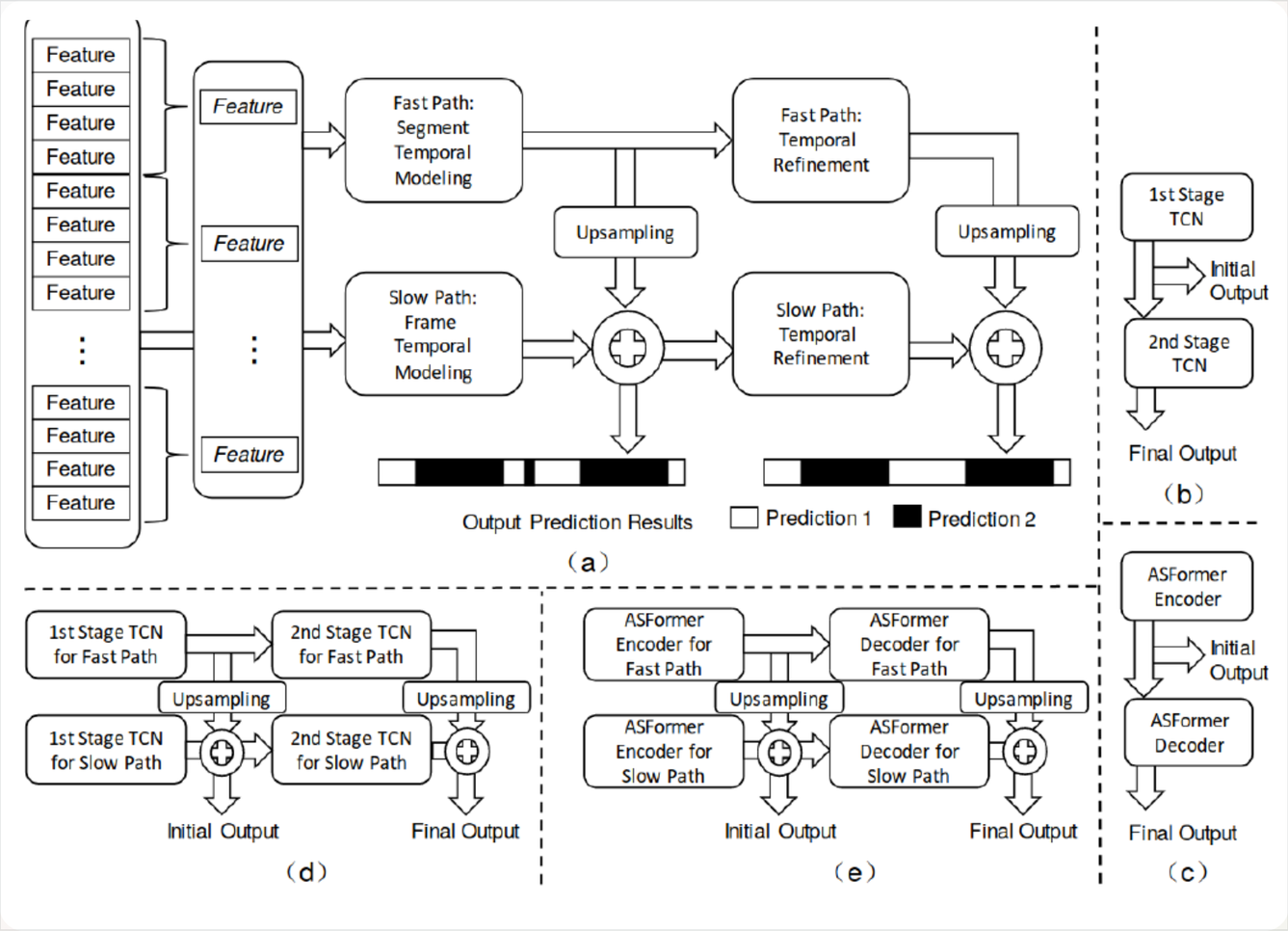
Phase + tools

Auxiliary instrument labels improve the shared representation.

Architecture figure: Abu Farha & Gall, CVPR 2019, Fig. 1

ARCHITECTURE DEEP DIVE

SF-TMN fuses frame and segment context



Slow · T positions

One prediction position per second preserves phase boundaries.

Fast · ≈ T / 32

32-second max pooling captures coarse workflow structure.

ASFormer blocks

Dilated convolution plus local attention at both resolutions.

Fusion + 3 refinements

Learned weights combine upsampled fast and slow features.

Architecture figure: Zhang et al., IJCARS 2024, Fig. 2

TRAINING OBJECTIVE

The loss must protect rare phases without creating flicker

Weighted CE

Raises the cost of errors on under-represented phases.

Focal loss

Focuses learning on difficult and low-confidence frames.

Boundary-aware

Emphasizes a five-second neighborhood around transitions.

Total loss = phase loss + 0.30 × smoothing loss + 0.40 × instrument loss

Feature augmentation: Gaussian noise and random feature dropout.

Optimization: early stopping on validation mean segmental F1.

Source: Project report, Sections 3.3-3.4

EVALUATION DISCIPLINE

Every choice is frozen before the test set is opened

01 · TRAIN

Fit candidates

Feature source × architecture × loss, seed 42.

02 · VALIDATE

Select once

Checkpoint, loss, raw vs Viterbi, tool threshold.

03 · TEST

Evaluate once

40 unseen videos; no further model selection.

Primary validation metric: mean of F1@10, F1@25, and F1@50.

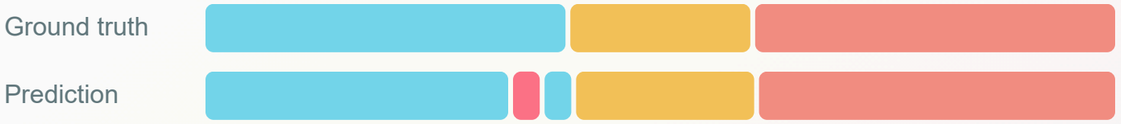
Source: Project report, Sections 3.4.3 and 4.6

WHAT COUNTS AS A GOOD PREDICTION?

Temporal metrics expose errors that accuracy can hide

Frame-level view

Accuracy, macro precision, macro recall, macro F1, and macro Jaccard.



Segment-level view

Edit penalizes extra or missing phases. **F1@k** matches predicted and real intervals by temporal IoU.

F1@10
lenient

F1@25
balanced

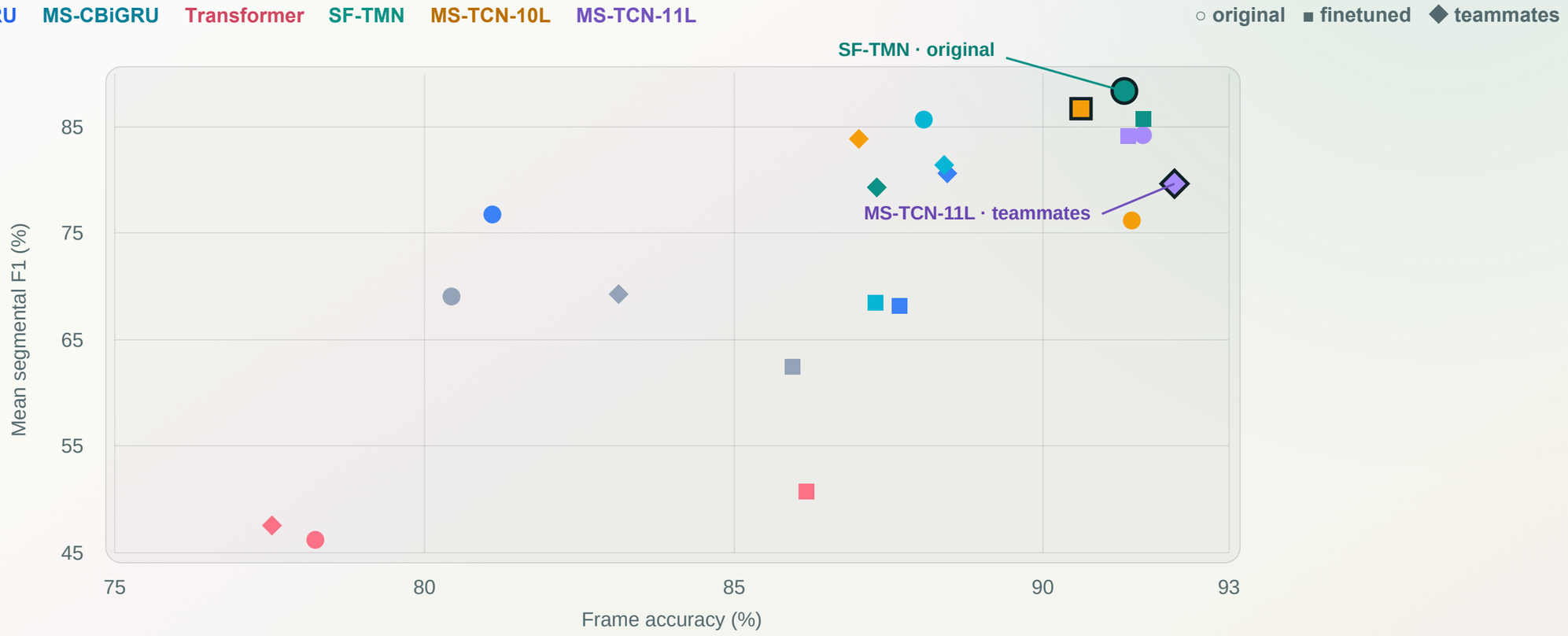
F1@50
strict

Viterbi is optional: validation keeps it only when it improves the temporal sequence.

Source: Project report, Chapter 4

THE CENTRAL FINDING

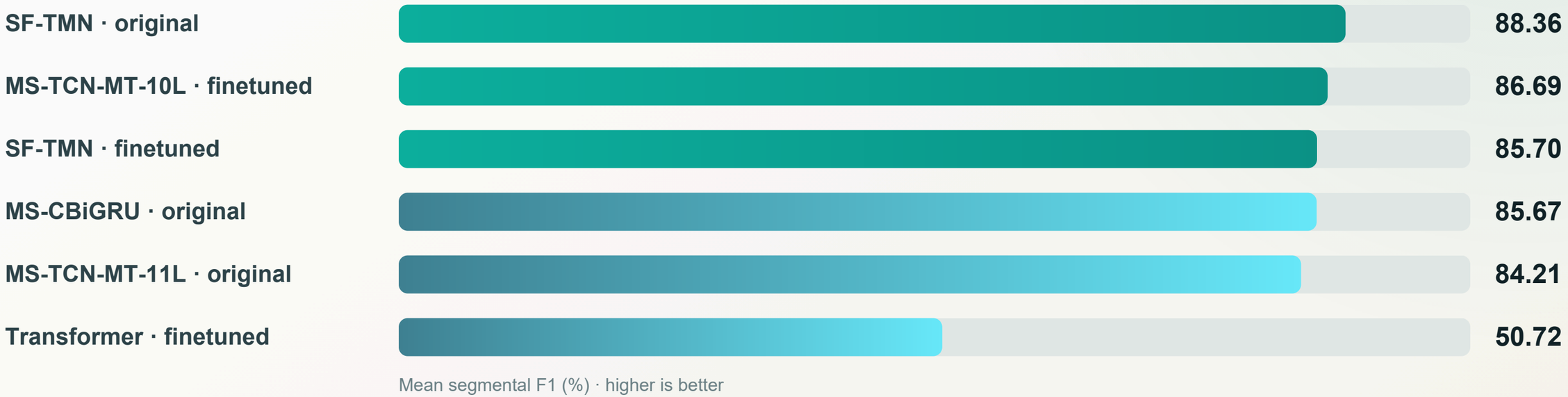
The accuracy winner is not the temporal winner



Source: Project report, Tables 5.1-5.2

PRIMARY METRIC RANKING

Full-sequence and multi-scale models dominate temporal quality



Source: Project report, Table 5.2

WHAT THE AVERAGES CONCEAL

Feature source and depth interact with architecture

Change in mean segmental F1 when the multi-task MS-TCN moves from **10 to 11 layers**:

original features

+8.01

76.20 → 84.21

finetuned features

−2.54

86.69 → 84.15

teammates features

−4.21

83.86 → 79.65

< 2.3 M

parameters in every model

< 9 s

test inference for 40 videos

53.60%

best instrument macro F1

Source: Project report, Sections 5.4-5.7

CONCLUSION

A surgical model is useful only when its sequence is clinically plausible.

Select SF-TMN original

Best workflow reconstruction: 88.36% mean segmental F1.

Keep metrics aligned

Accuracy alone can reward fragmented predictions.

Validate the next step

Multi-seed tests, controlled 10-vs-12 layers, and causal inference.

Questions

Source: Project report, Chapter 6

Frame-level metrics on the test set

Feature	Model	Post	Acc.	Prec. M	Rec. M	F1 M	Jacc. M
original	GRU	Vit.	80.44	76.48	83.45	78.31	65.01
original	BiGRU	Vit.	81.10	73.30	86.77	77.16	63.76
original	MS-CBiGRU	Vit.	88.07	81.15	89.11	83.75	72.83
original	Transformer	Vit.	78.24	76.54	70.87	72.21	57.01
original	SF-TMN	raw	91.31	85.89	88.02	86.84	77.18
original	MS-TCN-MT-10L	raw	91.43	86.30	88.09	87.10	77.52
original	MS-TCN-MT-11L	Vit.	91.61	**88.37**	85.79	86.80	77.15
finetuned	GRU	Vit.	85.95	77.61	82.44	79.45	66.48
finetuned	BiGRU	Vit.	87.68	79.07	84.32	81.08	68.90
finetuned	MS-CBiGRU	Vit.	87.29	78.22	84.69	80.74	68.44
finetuned	Transformer	Vit.	86.17	82.70	80.49	80.68	68.02
finetuned	SF-TMN	Vit.	91.62	88.08	87.57	87.73	78.36
finetuned	MS-TCN-MT-10L	Vit.	90.61	84.00	**89.92**	86.37	76.31
finetuned	MS-TCN-MT-11L	Vit.	91.37	87.41	88.67	**87.86**	**78.61**
teammates	GRU	Vit.	83.14	78.34	81.93	79.48	66.58
teammates	BiGRU	Vit.	88.45	79.73	86.71	81.99	70.23
teammates	MS-CBiGRU	Vit.	88.40	79.90	87.66	82.80	71.37
teammates	Transformer	Vit.	77.54	72.03	74.54	71.35	56.36
teammates	SF-TMN	raw	87.31	79.74	78.46	78.89	66.61
teammates	MS-TCN-MT-10L	Vit.	87.02	84.87	83.09	83.62	72.18
teammates	MS-TCN-MT-11L	raw	**92.12**	87.16	87.70	87.34	77.98

Source: Project report, Table 5.1 · values in percent

Temporal segmentation metrics on the test set

Feature	Model	Post	Edit	F1@10	F1@25	F1@50	Mean seg. F1
original	GRU	Vit.	65.06	72.99	71.55	62.64	69.06
original	BiGRU	Vit.	79.09	83.62	79.25	67.39	76.76
original	MS-CBiGRU	Vit.	85.71	89.67	87.67	79.67	85.67
original	Transformer	Vit.	44.63	52.49	47.41	38.73	46.21
original	SF-TMN	raw	**86.69**	**91.40**	**91.06**	**82.63**	**88.36**
original	MS-TCN-MT-10L	raw	77.22	79.35	78.17	71.09	76.20
original	MS-TCN-MT-11L	Vit.	81.78	87.17	86.51	78.95	84.21
finetuned	GRU	Vit.	57.21	67.08	64.36	55.94	62.46
finetuned	BiGRU	Vit.	64.36	72.68	70.79	61.10	68.19
finetuned	MS-CBiGRU	Vit.	63.85	72.41	71.06	61.91	68.46
finetuned	Transformer	Vit.	47.87	56.24	51.84	44.07	50.72
finetuned	SF-TMN	Vit.	85.73	89.11	87.79	80.20	85.70
finetuned	MS-TCN-MT-10L	Vit.	85.69	89.44	88.12	82.51	86.69
finetuned	MS-TCN-MT-11L	Vit.	84.26	86.97	86.32	79.15	84.15
teammates	GRU	Vit.	65.70	73.62	72.46	61.74	69.28
teammates	BiGRU	Vit.	79.71	84.98	83.39	73.48	80.62
teammates	MS-CBiGRU	Vit.	82.06	85.85	83.90	74.47	81.41
teammates	Transformer	Vit.	48.10	53.93	50.53	38.22	47.56
teammates	SF-TMN	raw	81.81	85.01	82.04	70.84	79.30
teammates	MS-TCN-MT-10L	Vit.	82.43	87.60	86.61	77.36	83.86
teammates	MS-TCN-MT-11L	raw	81.65	81.90	81.29	75.77	79.65

Source: Project report, Table 5.2 · values in percent

Auxiliary task, exploratory ablation, and efficiency

Instrument recognition

Feature / layer	Thr.	F1 M	F1 micro	Exact
original / 10L	0.30	**53.60**	70.09	32.13
original / 11L	0.30	49.05	69.90	31.49
finetuned / 10L	0.30	31.20	64.05	23.91
finetuned / 11L	0.40	48.16	**71.22**	**34.34**
teammates / 10L	0.30	50.02	69.33	31.12
teammates / 11L	0.30	50.16	69.87	31.67

Single-task MS-TCN exploratory ablation

Loss	Acc.	F1 M	Jacc. M	Edit
Weighted CE	91.32	**87.81**	**78.49**	79.13
Focal γ=1	91.43	87.36	77.80	**82.41**
Boundary-aware	**91.49**	87.00	77.31	81.15

Computational efficiency

Model	Params	Train min	VRAM MB	Test s
GRU	1,316,359	2.14-3.64	205-206	2.26-2.32
BiGRU	1,119,751	0.47-0.82	252	0.71-0.75
MS-CBiGRU	1,382,919	0.51-0.92	283	0.68-0.75
Transformer	2,173,959	6.99-16.72	144	0.96-1.04
SF-TMN	2,259,932	17.25-29.14	336	5.02-8.46
MS-TCN-MT-10L	796,792	0.78-2.10	158-159	0.67-2.18
MS-TCN-MT-11L	862,840	1.19-1.89	166	0.69-2.27

Interpretation

The instrument task remains imbalanced; the single-task loss comparison is exploratory because standardized segmental metrics were not exported.

Source: Project report, Tables 5.3-5.5 · values in percent unless stated